Statistics formula sheet

Northfield Institute - demo learning resource for the Lectern LMS template

Describing data

Mean = sum(x) / n Median = middle value when sorted

Variance s^2 = sum((x - mean)^2) / (n - 1) SD = sqrt(variance)

Probability

P(A or B) = P(A) + P(B) - P(A and B)

P(A and B) = P(A) x P(B) (independent events)

Normal distribution

z = (x - mean) / SD 68% within 1 SD, 95% within 2, 99.7% within 3

Inference

95% CI for a mean: mean +/- 1.96 x SD / sqrt(n) (large samples)

Two-proportion z test: z = (p1 - p2) / sqrt(p(1 - p)(1/n1 + 1/n2))